

1. What is First-Order Logic?

- First-Order Logic (FOL) is a more expressive knowledge representation language than propositional logic.
- Allows us to talk about objects, their properties, relations between objects, and quantification over objects.
- Uses predicates, functions, constants and variables to describe the world.
- Basis for logical inference in AI.

Key Idea: FOL extends propositional logic by allowing internal structure within sentences.

2. Why FOL is useful

- Represents specific objects (e.g. John, Paris).
- Expresses properties of objects (e.g. Human(John)).
- Expresses relations between objects (e.g. Brother(John, Richard)).
- Can state general rules using variables and quantifiers (for all / there exists).
- Useful for modeling complex real-world domains such as family relations, planning, knowledge about physical objects, etc.

FOL lets us represent what objects exist, what they are, and how they are related.

3. Basic Building Blocks

- Constants**
 - Name a specific object.
 - Start with an uppercase letter.
 - Examples: John, Paris, 3, A, UCB.
- Predicates**
 - Represent properties or relations.
 - Start with an uppercase letter.
 - Examples: Human(x), Loves(x, y), King(x), On(x, y).
- Functions**
 - Map objects to objects.
 - Start with an uppercase letter.
 - Examples: MotherOf(x), Sqrt(x), Location(x).
- Variables**
 - Range over objects in the domain.
 - Start with a lowercase letter.
 - Examples: x, y, z.

4. Atomic Sentences

- An atomic sentence (or atom) is a predicate with a list of terms as arguments.
- General form: $P(t_1, t_2, \dots, t_n)$
- Represents a basic fact or relation in the world.
- It is either true or false in a particular model (interpretation).

Examples: Human(John)
Loves(John, Mary)
Sibling(Ann, Bob)

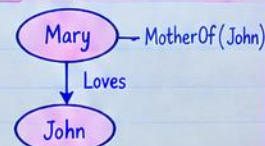
5. Terms in FOL

- Terms are expressions that refer to objects.
- (a) Constants** – refer to a specific object.
Examples: John, Paris, 3, UCB.
- (b) Variables** – range over objects in the domain.
Examples: x, y, z.
- (c) Function terms** – denote objects, constructed using function symbols and terms.
 - If f is an n-ary function and $t_1, \dots, t_n$ are terms, then $f(t_1, \dots, t_n)$ is a term.
 - Can be nested: MotherOf(FatherOf(x)).
 - Examples: MotherOf(John), Sqrt(4), Location(Paris), MotherOf(MotherOf(x)).

6. Example Facts in FOL

- King(John) (John is a king)
- Greedy(John) (John is greedy)
- Brother(Richard, John) (Richard is a brother of John)
- Loves(MotherOf(John), John) (John's mother loves John)

Example: Family relations as facts



7. Real-World Examples of FOL Representations

(a) Family Domain

Objects: people in a family.
Predicates: Parent(x, y), Father(x, y), Mother(x, y), Brother(x, y), Sister(x, y), Sibling(x, y), Ancestor(x, y).
Functions: MotherOf(x), FatherOf(x).
Example facts:
 Father(John, Mary)
 Mother(Mary, Alice)
 Sibling(Bob, Mary)
 Ancestor(John, Alice)

(b) University Domain

Objects: people, courses, departments.
Predicates: Student(x), Professor(x), Course(x), Department(x), Teaches(x, y), Takes(x, y), Advisor(x, y).
Functions: AdvisorOf(x), DepartmentOf(x).
Example facts:
 Student(Alice)
 Professor(Brown)
 Teaches(Brown, CS101)
 Takes(Alice, CS101)
 AdvisorOf(Alice, Brown)

(c) Geography Domain

Objects: cities, countries, physical locations.
Predicates: City(x), Country(x), On(x, y), Border(x, y), River(x), LocatedIn(x, y).
Functions: CapitalOf(x), Length(x).
Example facts:
 City(Paris)
 Country(France)
 CapitalOf(France, Paris)
 LocatedIn(Paris, France)
 River(Nile)
 Length(Nile, 6650)



8. FOL vs Propositional Logic

Aspect	Propositional Logic	First-Order Logic
Basic unit	Atomic proposition (P, Q, ...)	Predicate with terms (P(x), R(x, y))
Expressiveness	Very limited	More expressive
Objects	No explicit objects	Represents specific objects
Relations	Not directly represented	Represents relations between objects
Quantification	Not available	Supports $\forall$ (for all) and $\exists$ (there exists)
Example	P, Q, $P \wedge Q$	Human(John), $\forall x \text{ Human}(x) \rightarrow \text{Mortal}(x)$

9. Key Idea / Takeaway

- First-Order Logic allows us to represent and reason about objects, their properties, relations, and general rules.
- Uses constants, predicates, functions and variables to build complex sentences.
- Forms the basis for logical inference and knowledge representation in AI.
- Much more expressive than propositional logic, enabling modeling of real-world domains.

FOL is a powerful language for describing the structure of the world.

10. Summary Quote

“First-Order Logic gives us the language to talk about objects, their relationships, and the rules that govern the world.”

– AIMA (Ch. 8)

1. Syntax of First-Order Logic (FOL)

- FOL has a well-defined syntax for building valid sentences.
- Sentences are built from terms, predicates, connectives and quantifiers.
- Well-formed formulas (wffs) are defined recursively.
- Every FOL sentence is a wff.

Key Idea: FOL syntax specifies the legal form of expressions, independent of meaning (in semantics).

2. Predicates and Terms

- Predicates represent relations over objects.
- A predicate takes terms as arguments and returns a truth value.
- Terms are used to refer to objects.

Types of terms:

- Constants:** refer to specific objects (John, Paris, 3).
- Variables:** range over objects (x, y, z).
- Function terms:** denote objects constructed from other terms (e.g. $\text{FatherOf}(x)$):

Examples of predicates (with arity)

Human(x)	(1)
Loves(x, y)	(2)
Sibling(x, y)	(2)
Greater(x, y)	(2)

Examples of terms

John (constant)
x, y, z (variables)
MotherOf(x) (function)
Location(Paris) (function)
3 (constant)

3. Logical Connectives

- FOL uses the same logical connectives as propositional logic.

Connective	Meaning	Notation
Negation	not	$\neg \phi$
Conjunction	and	$\phi \wedge \psi$
Disjunction	or	$\phi \vee \psi$
Implication	if... then	$\phi \rightarrow \psi$
Biconditional	iff	$\phi \leftrightarrow \psi$

Example:

- $\text{Human}(\text{John}) \wedge \text{Mortal}(\text{John})$
- $\neg \text{Cat}(\text{John})$
- $\text{Loves}(x, y) \rightarrow \text{Happy}(x)$
- $P(x) \leftrightarrow Q(x)$

4. Quantifiers in FOL

- FOL has two quantifiers: universal ($\forall$) and existential ($\exists$).
- They specify how many objects in the domain make a formula true.
- Variables in a sentence can be quantified or free (unquantified).

Key Idea: Quantifiers allow us to make general statements about all or some objects in the domain.

5. Universal Quantifier ($\forall$)

- $\forall x P(x)$ means $P(x)$ is true for all objects x in the domain.
- It makes a general statement about every object.

Example: $\forall x \text{Human}(x) \rightarrow \text{Mortal}(x)$
"All humans are mortal."

Other examples:

- $\forall x \text{Cat}(x) \rightarrow \text{Mammal}(x)$
"All cats are mammals."
- $\forall x \forall y \text{Loves}(x, y) \rightarrow \text{Likes}(y, x)$
"If someone loves someone, they like them."

6. Existential Quantifier ($\exists$)

- $\exists x P(x)$ means $P(x)$ is true for some object x in the domain.
- It states that there exists at least one object with the property.

Example: $\exists x \text{Cat}(x) \wedge \text{Black}(x)$
"There exists a black cat."

Other examples:

- $\exists x \text{Human}(x) \wedge \text{Tall}(x)$
"There is a tall human."
- $\exists x \text{Likes}(\text{John}, x)$
"John likes someone."

7. Scope of Quantifiers

- A quantifier applies to the formula that follows it (its scope).
- Parentheses are used to make scope clear.
- Quantifiers can have nested scope.

Example: $\forall x (P(x) \rightarrow Q(x))$
means "for all x , if $P(x)$ then $Q(x)$ "
 $(\forall x P(x)) \rightarrow Q(x)$
means "if everything is P , then Q ."

Nested quantifiers:

$\forall x \exists y \text{Loves}(x, y)$
"For every x , there exists a y such that x loves y ."

8. Examples of Quantified Sentences

- $\forall x \text{Human}(x) \rightarrow \text{Mortal}(x)$
"All humans are mortal."
- $\exists x \text{Cat}(x) \wedge \text{Black}(x)$
"There exists a black cat."
- $\forall x \exists y \text{Loves}(x, y)$
"Everyone loves someone."
- $\exists x \forall y \text{Larger}(x, y)$
"There exists something that is larger than everything."
- $\forall x (\text{Parent}(x, y) \rightarrow \text{Ancestor}(x, y))$
"For all x , if x is a parent of y then x is an ancestor of y ."

9. Translating English Sentences into FOL

- Identify the main predicates, constants and variables.
- Determine the quantifiers.
- Be careful about the scope and meaning.

English Sentence	FOL Translation
All humans are mortal.	$\forall x \text{Human}(x) \rightarrow \text{Mortal}(x)$
There exists a black cat.	$\exists x \text{Cat}(x) \wedge \text{Black}(x)$
Everyone loves someone.	$\forall x \exists y \text{Loves}(x, y)$
Someone is loved by everyone.	$\exists x \forall y \text{Loves}(y, x)$
No cats are dogs.	$\forall x \neg (\text{Cat}(x) \wedge \text{Dog}(x))$
Every student has a friend.	$\forall x \text{Student}(x) \rightarrow \exists y \text{Friend}(x, y)$
There is a person in Paris.	$\exists x \text{Person}(x) \wedge \text{LocatedIn}(x, \text{Paris})$

Note: The same English sentence can sometimes be translated in different (but related) ways, depending on the intended meaning.

10. Summary and Key Takeaways

- FOL syntax uses terms, predicates, connectives and quantifiers.
- Predicates and functions allow us to talk about objects and their relations.
- $\forall$ means "for all" and $\exists$ means "there exists".
- Quantifier scope is important and parentheses remove ambiguity.
- FOL lets us represent general knowledge about the world.
- Translating English sentences into FOL helps formalize natural language statements.

Key Takeaways:

- FOL has a precise syntax for building sentences.
- Predicates and terms refer to objects and relations.
- Quantifiers make general statements about objects.
- Scope determines which part of the formula is quantified.
- FOL is a powerful and expressive representation language used throughout AI.

1. Semantics of First-Order Logic

- The meaning (semantics) of a FOL sentence is defined with respect to a model (interpretation).
- Semantics tells us when a sentence is true in a particular world (model).
- Unlike propositional logic, truth depends on the domain of objects and the interpretation of symbols (predicates, functions, constants).

Key Idea: The meaning of a FOL sentence depends on the model and interpretation that we use.

2. Interpretations (Models)

- An interpretation (model) specifies:
 - A domain of objects (non-empty set).
 - An interpretation of constant symbols (as objects in the domain).
 - An interpretation of predicate symbols (as relations over objects).
 - An interpretation of function symbols (as functions over objects).

A model provides a possible world in which the sentence can be evaluated as true or false.

3. Domain of Discourse

- The domain (universe of discourse) is the set of objects that variables can refer to.
- The domain can be finite or infinite.
- Constants, functions and predicates are interpreted with respect to this domain.

Example: Domain of people

Domain $D = \{\text{John, Mary, Bob}\}$

Constant: $\text{John} \rightarrow \text{John}$

Predicate: $\text{Human}(x) \rightarrow \text{set of human objects}$

Function: $\text{Father}(x) \rightarrow \text{a person (in } D)$

4. Truth in a Model

- A sentence is true in a model if it is satisfied by the interpretation in that model.
- The truth of atomic sentences is determined by the interpretation of predicates, functions and constants.
- The truth of complex sentences is defined recursively using the usual logical connectives and quantifiers.

Notation: $M \models \alpha$ means "sentence α is true in model M " (M satisfies α).

5. Example: A Model for a FOL Sentence

Consider the sentence: $\forall x (\text{Human}(x) \rightarrow \text{Mortal}(x))$

Model M:

- Domain: $D = \{\text{John, Mary}\}$
- $\text{Human}(x)$: $\{\text{John, Mary}\}$
- $\text{Mortal}(x)$: $\{\text{John, Mary}\}$

Is the sentence true?

- For $x = \text{John}$: $\text{Human}(\text{John})$ is true and $\text{Mortal}(\text{John})$ is true.
- For $x = \text{Mary}$: $\text{Human}(\text{Mary})$ is true and $\text{Mortal}(\text{Mary})$ is true.
- Therefore, $M \models \forall x (\text{Human}(x) \rightarrow \text{Mortal}(x))$ (the sentence is true in this model).

6. Satisfiable, Valid and Unsatisfiable Sentences

Type	Meaning	Symbol	Example
Satisfiable	True in at least one model.	$\exists \text{ model } M \text{ such that } M \models \alpha$	$\exists x \text{ Human}(x)$ (true if domain has at least one human)
Valid (tautology)	True in every model.	$\forall \text{ models } M, M \models \alpha$	$\forall x (\text{Human}(x) \rightarrow \text{Mortal}(x))$ (in intended domain of people)
Unsatisfiable (contradiction)	False in every model.	$\forall \text{ models } M, M \not\models \alpha$	$\exists x (\text{Human}(x) \wedge \neg \text{Human}(x))$

7. Models: More Examples

(a) A model where a sentence is true

Domain: $D = \{\text{Alice, Bob}\}$
 $\text{Human}(x)$: $\{\text{Alice}\}$
 $\text{Loves}(x, y)$: $\{(\text{Alice, Bob})\}$

In this model:

- $\text{Human}(\text{Alice})$ is true.
- $\text{Human}(\text{Bob})$ is false.
- $\text{Loves}(\text{Alice, Bob})$ is true.

Example sentences:

- $\text{Human}(\text{Alice}) \rightarrow \text{true}$
- $\text{Human}(\text{Bob}) \rightarrow \text{false}$
- $\exists x \text{ Human}(x) \rightarrow \text{true}$
- $\forall x \text{ Human}(x) \rightarrow \text{false}$
- $\text{Loves}(\text{Alice, Bob}) \rightarrow \text{true}$
- $\text{Loves}(\text{Bob, Alice}) \rightarrow \text{false}$

(b) A model where a sentence is false

Domain: $D = \{\text{John}\}$
 $\text{Human}(x)$: $\{\text{John}\}$
 $\text{Mortal}(x)$: $\emptyset$ (nobody is mortal)

Sentence:

$\forall x (\text{Human}(x) \rightarrow \text{Mortal}(x))$

In this model: **false**

8. Entailment in First-Order Logic

- A sentence α entails a sentence β (written $\alpha \models \beta$) if in every model where α is true, β is also true.
- Entailment is a semantic (model-theoretic) notion.
- $\alpha \models \beta$ means: for all models M , if $M \models \alpha$ then $M \models \beta$.
- Entailment captures logical consequence between sentences.

Key Idea: $\alpha \models \beta$ means " α being true guarantees that β is true, in all possible models".

Examples:

- $\forall x \text{ Human}(x) \models \exists x \text{ Human}(x)$
(if everyone is human, then someone is human).
- $\forall x (\text{Human}(x) \rightarrow \text{Mortal}(x)), \text{Human}(\text{John}) \models \text{Mortal}(\text{John})$
(from a general rule and a fact, we can infer a specific fact).

9. More Examples: Satisfiability, Validity and Entailment

(a) Satisfiable sentence

- α : $\exists x \text{ Loves}(x, x)$
- True in a model where someone loves themselves (e.g., domain $\{\text{Alice}\}$ with $\text{Loves}(\text{Alice, Alice})$).
 - So α is satisfiable (but not valid).

(b) Valid sentence

- β : $\forall x (\text{P}(x) \vee \neg \text{P}(x))$
- True in every model (law of excluded middle).
 - So β is valid.

(c) Unsatisfiable sentence

- γ : $\forall x (\text{P}(x) \wedge \neg \text{P}(x))$
- False in every model.
 - So γ is unsatisfiable.

Entailment example: α : $\forall x (\text{Human}(x) \rightarrow \text{Mortal}(x))$, $\text{Human}(\text{John})$
 β : $\text{Mortal}(\text{John})$

Then $\alpha \models \beta$ (in every model, if α is true, β is true).

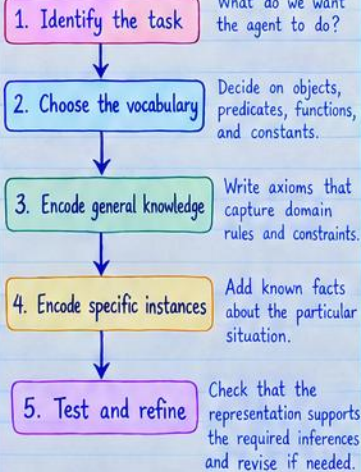
"The meaning of a FOL sentence is defined relative to a model, and entailment holds in all models." — AIMA (Ch.8)

1. Using FOL for Knowledge Representation

- FOL is a knowledge representation language for describing objects, their properties, relations, and general rules about a domain.
- Allows us to represent facts and constraints about the world in a clear, declarative way.
- Supports reasoning to derive new facts from existing knowledge.
- Can represent general statements using variables and quantifiers.
- Widely used in AI for domains such as natural language, planning, vision, and robotics.

Key Idea: FOL provides a rich, expressive way to represent structured knowledge about the world.

2. Knowledge Engineering Process



3. Choosing Objects, Predicates, Functions

(a) Objects (Constants)

- Represent specific individuals in the domain.
- Choose constants for important named objects.
- Examples: John, Mary, Paris, CS101, A, B.

(b) Predicates

- Represent properties or relations among objects.
- Choose predicates that are relevant to the task.
- Use clear, intuitive names and correct arity.
- Examples: $\text{Parent}(x, y)$, $\text{City}(x)$, $\text{On}(x, y)$, $\text{Student}(x)$, $\text{NorthOf}(x, y)$.

(c) Functions

- Represent objects constructed from other objects.
- Choose functions when objects have natural structure or parts.
- Examples: $\text{MotherOf}(x)$, $\text{FatherOf}(x)$, $\text{CapitalOf}(x)$, $\text{Left}(x)$, $\text{Right}(x)$.

Good representations use as few symbols as necessary, but enough to express the required knowledge clearly.

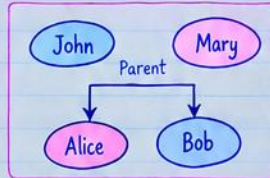
4. Family Domain Example

Objects: people in a family.

Constants: John, Mary, Alice, Bob.

Example facts:

$\text{Parent}(\text{John}, \text{Alice})$
 $\text{Parent}(\text{Mary}, \text{Alice})$
 $\text{Parent}(\text{John}, \text{Bob})$
 $\text{Parent}(\text{Mary}, \text{Bob})$
 $\text{Male}(\text{John})$
 $\text{Female}(\text{Mary})$



Example rule:

$\forall x \forall y (\text{Parent}(x, y) \rightarrow (\text{Mother}(y, x) \rightarrow \text{Female}(x)))$

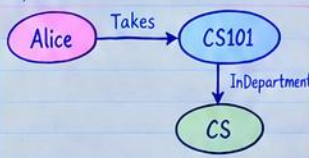
6. University Domain Example

Objects: people, courses, departments.

Constants: Alice, Bob, CS101, CS, AI.

Example facts:

$\text{Student}(\text{Alice})$
 $\text{Student}(\text{Bob})$
 $\text{Course}(\text{CS101})$
 $\text{Department}(\text{CS})$
 $\text{Major}(\text{Alice}, \text{CS})$
 $\text{Takes}(\text{Alice}, \text{CS101})$
 $\text{Takes}(\text{Bob}, \text{CS101})$
 $\text{Instructor}(\text{ProfSmith}, \text{CS101})$



Example rule:

$\forall x \forall y (\text{Takes}(x, y) \rightarrow \text{Student}(x) \wedge \text{Course}(y))$

5. Geography Domain Example

Objects: cities, countries, rivers, physical locations.

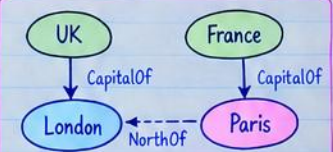
Constants: Paris, London, France, UK, Nile.

Example facts:

$\text{City}(\text{Paris})$
 $\text{City}(\text{London})$
 $\text{Country}(\text{France})$
 $\text{Country}(\text{UK})$
 $\text{CapitalOf}(\text{Paris}, \text{France})$
 $\text{CapitalOf}(\text{London}, \text{UK})$
 $\text{LocatedIn}(\text{Paris}, \text{France})$
 $\text{NorthOf}(\text{Paris}, \text{London})$

Example rule:

$\forall x \forall y (\text{CapitalOf}(x, y) \rightarrow \text{City}(x) \wedge \text{Country}(y))$



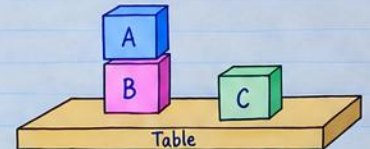
7. Blocks World Domain Example

Objects: blocks and the table.

Constants: A, B, C, Table.

Example facts:

$\text{Block}(A)$
 $\text{Block}(B)$
 $\text{Block}(C)$
 $\text{On}(A, B)$
 $\text{On}(B, \text{Table})$
 $\text{On}(C, \text{Table})$
 $\text{Clear}(A)$
 $\text{Clear}(C)$



Example rule:

$\forall x (\text{On}(x, \text{Table}) \rightarrow \text{Clear}(x) \rightarrow \text{Moveable}(x))$

8. Tips for Writing Good FOL Representations

- Keep it simple:**
 - Use the smallest set of predicates and functions needed.
 - Avoid unnecessary complexity.
- Be clear and consistent:**
 - Use intuitive names and consistent notation.
 - Specify the meaning (intended interpretation) of each symbol.
- Choose the right level of detail:**
 - Include enough detail to support the required inferences.
 - Do not model irrelevant details.
- Separate general knowledge and specific facts:**
 - Write general rules (axioms) separately from instance facts.
- Consider the domain's structure:**
 - Use types or predicate constraints to restrict the domain.
 - Exploit hierarchies (e.g., City is a type of Location).
- Test with example queries:**
 - Check that the representation can answer important questions.
 - Refine and add axioms as needed.

9. Common Modeling Choices and Examples

Aspect	Guidelines	Examples
Objects	Use constants for specific named individuals.	John, Paris, CS101, A
Predicates	Use predicates for relevant properties and relations.	$\text{Parent}(x, y)$, $\text{City}(x)$, $\text{On}(x, y)$, $\text{Student}(x)$
Functions	Use functions for object structure or unique related objects.	$\text{MotherOf}(x)$, $\text{CapitalOf}(x)$, $\text{Left}(x)$, $\text{Right}(x)$
Types / Domain constraints	Use unary predicates or types to restrict objects.	$\text{Person}(x)$, $\text{City}(x)$, $\text{Block}(x)$
Axioms / Rules	Write general rules with quantifiers.	$\forall x (\text{Student}(x) \rightarrow \text{Person}(x))$
Facts	Add specific instance facts about the current situation.	$\text{Student}(\text{Alice})$, $\text{On}(A, B)$, $\text{City}(\text{Paris})$

"A good knowledge representation captures the important aspects of the world, and is simple, clear, and supports the inferences we need."

1. Advantages of First-Order Logic

- More expressive than propositional logic.
- Can represent objects, properties, relations and quantification.
- Natural and intuitive representation of real-world knowledge.
- Easier to model complex domains (e.g. family, planning, spatial, time).
- Supports general rules (using variables).
- Allows inference about new, unseen objects (via quantifiers).

Key Idea: FOL can talk about a whole class of objects, not just specific facts.

2. Limitations of First-Order Logic

- Cannot naturally represent higher-order relations (e.g. relations about relations).
- Cannot directly express uncertainty, probabilities or degrees of belief.
- Cannot easily represent vague or fuzzy concepts (needs extensions).
- Time, actions and change require additional formalisms (e.g. situation calculus, temporal logic).
- Some real-world knowledge (e.g. defaults, exceptions) is difficult to express.
- Inference can be computationally expensive (undecidable in general).

3. FOL vs. Propositional Logic

Aspect	Propositional Logic	First-Order Logic
Basic unit	Atomic proposition (P, Q, ...)	Predicate with terms P(x), R(x, y)
Expressiveness	Less expressive	Much more expressive
Objects	No explicit objects	Represents objects
Relations	Not directly represented	Represents relations
Quantification	Not available	Uses $\forall$ and $\exists$
Domain	Implicit (no domain)	Explicit domain of discourse
Example	P, Q, $P \wedge Q$	Human(x), $\forall x (\text{Human}(x) \rightarrow \text{Mortal}(x))$

4. Example Comparison

Propositional Logic	First-Order Logic
Each fact is a separate symbol.	General rule using variables.
Example:	Example:
• P = "John is a human" • Q = "John is mortal" • $P \rightarrow Q$ = "If John is a human then John is mortal"	• Human(John) • Mortal(John) • $\forall x (\text{Human}(x) \rightarrow \text{Mortal}(x))$ From the rule, we can infer Mortal(John) for any human, not just John.

5. Common Pitfalls in FOL

(a) Confusing constants and variables

- Constants refer to specific objects.
 - Variables range over all objects.
- Example:** Human(John) ✓
Human(x) ✓ (x is a variable)

(b) Incorrect use of quantifiers

- $\forall x P(x)$ is not the same as $\exists x P(x)$.
 - Negation must be handled correctly.
- Example:** $\neg \forall x P(x) \equiv \exists x \neg P(x)$

(c) Free vs. bound variables

- Every variable in a sentence must be bound by a quantifier.
- Example:** Human(x) $\rightarrow$ Mortal(x)
(has a free variable, not a sentence)
 $\forall x (\text{Human}(x) \rightarrow \text{Mortal}(x))$
(is a sentence)

6. More Common Mistakes

(d) Misusing functions and predicates

- Functions return objects; predicates return truth values.
- Example:** Father(John) ✗ (wrong)
Father(John, y) ✓ (correct)

(e) Wrong scoping of quantifiers

- The scope of a quantifier includes everything to its right (unless grouped).
- Example:** $\forall x P(x) \wedge Q(x)$
means $(\forall x P(x)) \wedge Q(x)$
- Use parentheses: $\forall x (P(x) \wedge Q(x))$

(f) Assuming domain when not specified

- FOL assumes a non-empty domain but does not specify what it is.
- Always be clear about the domain in examples and models.

7. Sample FOL Sentences (Good vs. Bad)

Good Examples (Well-formed)	Bad Examples (Common Errors)
• Human(John) • Loves(John, Mary) • $\forall x (\text{Human}(x) \rightarrow \text{Mortal}(x))$ • $\exists x (\text{King}(x) \wedge \text{Greedy}(x))$ • Mother(Anna, x) • $\exists x (\text{Parent}(x, y) \wedge \text{Loves}(x, y))$	• Human() (missing argument) • Loves(John) (wrong arity) • $\forall (\text{Human}(x))$ (missing variable) • Human(x $\rightarrow$ y) (using $\rightarrow$ inside term) • $\forall x \text{Human}(x \rightarrow \text{Mortal}(x))$ (wrong scope) • $\exists x \forall y \text{Loves}(x, y)$ (missing parenthesis)

9. Key Takeaways for Chapter 8

- 1 FOL is a powerful and expressive language for representing knowledge about objects, their properties, relations and general rules.
- 2 It extends propositional logic by adding predicates, functions, constants, variables and quantifiers.
- 3 FOL allows more natural and concise representation of real-world domains.
- 4 It has limitations: no direct support for uncertainty, higher-order relations, or commonsense defaults.
- 5 Always be careful with quantifiers, scope, and the use of variables, constants and functions.
- 6 FOL is widely used as the basis for logical inference and knowledge representation in AI.

8. Sample Exam Points

- 1 Write FOL sentences for English statements.
 - e.g. "All birds can fly." $\rightarrow \forall x (\text{Bird}(x) \rightarrow \text{CanFly}(x))$
- 2 Translate between English and FOL.
 - e.g. "There exists a student who likes all courses." $\rightarrow \exists x (\text{Student}(x) \wedge \forall y (\text{Course}(y) \rightarrow \text{Likes}(x, y)))$
- 3 Identify free and bound variables.
- 4 Convert sentences using quantifier equivalences.
 - e.g. $\neg \forall x P(x) \equiv \exists x \neg P(x)$
- 5 Compare FOL with propositional logic (advantages, expressiveness).
- 6 Identify errors in FOL formulas.
- 7 Short questions on functions, predicates, constants and domain.
- 8 Conceptual questions on limitations and use cases.

"First-Order Logic gives us the language to talk about a rich world of objects, their properties and relations, but we must use it carefully and understand its limits."